IV.6.1 Positive Definite Matrices

Recall that a square matrix with $A^T = A$ is called *symmetric*.

e.g.

$$A = \begin{bmatrix} 1 & -2 & 4 \\ -2 & 0 & \pi \\ 4 & \pi & 1.5 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -2 & e & \pi^2 & 4 \\ e & 2 & \sqrt{2} & \sin 1 \\ \pi^2 & \sqrt{2} & 1 & 2 \\ 4 & \sin 1 & 2 & 0 \end{bmatrix}$$

are symmetric matrices.

$A = A^T$

Definition: A <u>symmetric</u> $n \times n$ matrix A is called

a) *positive definite*, if for <u>every</u> (column) vector $\mathbf{x} \in \mathbb{R}^n$ with $\mathbf{x} \neq 0$,

$$\mathbf{x}^T A \mathbf{x} > 0$$

+

this is a real number

b) *negative definite*, if for <u>every</u> (column) vector $\mathbf{x} \in \mathbb{R}^n$ with $\mathbf{x} \neq 0$,

$$\mathbf{x}^T A \mathbf{x} < 0$$

−

How to check whether a symmetric matrix is positive / negative definite:

Case 1: A diagonal matrix A.

Shortcut notation: all entries above the diagonal are zero.

A square matrix of the form

$$A = \begin{bmatrix} a_1 & 0 & 0 & 0 & 0 \\ 0 & a_2 & 0 & 0 & 0 \\ 0 & 0 & a_3 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & a_n \end{bmatrix} \quad \text{(or simply } A = \begin{bmatrix} a_1 & & & & (0) \\ & a_2 & & & \\ & & a_3 & & \\ & & & \ddots & \\ (0) & & & & a_n \end{bmatrix}\text{)}$$

is called a *diagonal* matrix.

Shortcut notation: All entries below the diagonal are zero.

Example 5: Let A be a 3×3 diagonal matrix,

$$A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}.$$

diagonal matrix

For any vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ we have

(at least one x_i is $\neq 0$)

$$\mathbf{x}^T A\mathbf{x} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \underbrace{\begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\text{Multiply}} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}\begin{bmatrix} ax_1 \\ bx_2 \\ cx_3 \end{bmatrix} = \underset{\geq 0}{ax_1^2} + \underset{\geq 0}{bx_2^2} + \underset{\geq 0}{cx_3^2}$$

We observe:

-- $\mathbf{x}^T A\mathbf{x}$ is $\overset{\text{positive definite}}{>} 0$ for **all** x_1, x_2, x_3 (where at least one $x_i \neq 0$)

precisely when $\boxed{a > 0,\ b > 0,\ c > 0.}$

-- $\mathbf{x}^T A\mathbf{x}$ is $\overset{\text{negative definite}}{<} 0$ for **all** x_1, x_2, x_3 (where at least one $x_i \neq 0$)

precisely when $\boxed{a < 0,\ b < 0,\ c < 0.}$

Remark: The same idea can be applied to an arbitrary $n \times n$ <u>diagonal matrix</u>

$$A = \begin{bmatrix} a_1 & & & & (0) \\ & a_2 & & & \\ & & a_3 & & \\ & & & \ddots & \\ (0) & & & & a_n \end{bmatrix}$$

--- A is <u>positive definite</u>, precisely if **all** $a_i > 0$.

--- A is <u>negative definite</u>, precisely if **all** $a_i < 0$.

e.g. $$A = \begin{bmatrix} 1 & & & (0) \\ & 4 & & \\ & & \pi & \\ (0) & & & 10^{-9} \end{bmatrix}$$ is positive definite .

(1) diagonal matrix
(2) all entries in the diagonal are (+)
$\Downarrow$
pos. definite

Case 2: A general symmetric matrix A.

1. *(Necessary condition)*

--- If A is positive definite, then **all** entries in the diagonal are >0.

--- If A is negative definite, then **all** its entries in the diagonal are <0.

(i) Careful: when all entries in the diagonal are >0, we cannot yet conclude that A is positive definite !

(ii) The above are simple criteria to check that a matrix is NOT pos./neg. definite)

Example 10: Are the following matrices positive or negative definite ?

$$A = \begin{bmatrix} 4 & 5 & 2 & 4 \\ 5 & -3 & 2.6 & 1 \\ 2 & 2.6 & -2 & 0.7 \\ 4 & 1 & 0.7 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} 2 & -1 & 0.42 \\ -1 & 4 & -2 \\ 0.42 & -2 & 0 \end{bmatrix} \qquad C = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 4 & -2 \\ 0 & -2 & 7 \end{bmatrix}$$

Sol:

Matrix A: The diagonal contains positive (⊕) and negative (⊖) entries

$\Rightarrow$ A cannot be pos. definite, and cannot be neg. definite.

Matrix B: The diagonal contains a 0.

$\Rightarrow$ A cannot be pos. definite

Matrix C: The entries in the diagonal are all ≥ 0. So C is possibly pos. definite: We must check further as follows:

The criterion to check whether matrix C is positive definite is as follows:
We compute all sub-determinants that start in the top left corner:

$$C = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 4 & -2 \\ 0 & -2 & 7 \end{bmatrix}$$

$\boxed{D_1} = 2 > 0$ (+) (determinant of top left corner)

$$D_2 = \begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} = (4)(2) - (1)(1) = 7 > 0 \ (+)$$

(determinant of 2×2 top left corner)

$$D_3 = \begin{vmatrix} 2 & 1 & 0 \\ 1 & 4 & -2 \\ 0 & -2 & 7 \end{vmatrix} = \left|\begin{array}{ccc|cc} 2 & 1 & 0 & 2 & 1 \\ 1 & 4 & -2 & 1 & 4 \\ 0 & -2 & 7 & 0 & -2 \end{array}\right| = 41 > 0 \ (+)$$

(determinant of 3×3 top left corner, i.e. of C)

Because all these sub-determinants are >0, matrix C is positive definite.

2. *(Sylvester Criterion: Sufficient condition)*

--- matrix A is <u>positive definite</u> precicsely when <u>all sub-determinants</u> that start at the top left corner are > 0 (i.e. all have $+$ sign)

--- matrix A is <u>negative definite</u> precicsely when <u>all sub-determinants</u> that start at the top left corner have *alternating* signs of the form

$$-, +, -, +, \ldots$$

Example 11: Is the matrix $A = \begin{bmatrix} -2 & 1 & 0 \\ 1 & -4 & -2 \\ 0 & -2 & -8 \end{bmatrix}$ negative definite ?

Sol: Because all diagonal entries are $-$, the matrix A may possibly be negative definite. Let us check the sub-determinants to make sure :

$D_1 = -2 < 0$ $-$ (determinant of top left corner)

$D_2 = \begin{vmatrix} -2 & 1 \\ 1 & -4 \end{vmatrix} = (-2)(-4) - (1)(1) = 7 > 0$ $+$ (determinant of 2×2 top left corner)

$$D_3 = \begin{vmatrix} -2 & 1 & 0 \\ 1 & -4 & -2 \\ 0 & -2 & -8 \end{vmatrix} = \begin{vmatrix} -2 & 1 & 0 & -2 & 1 \\ 1 & -4 & -2 & 1 & -4 \\ 0 & -2 & -8 & 0 & -2 \end{vmatrix} = -48 < 0$$

$-$

(determinant of 3×3 top left corner, i.e. of A)

The signs of the sub-determinants D_1 , D_2, D_3 follow the pattern $-,+,-$:

$\Rightarrow$ A is negative definite.

Example 6: Is this matrix positive definite ?

$$A = \begin{bmatrix} 2 & 0 & 3 & 0 \\ 0 & 4 & 1 & 0 \\ 3 & 1 & 2 & 3 \\ 0 & 0 & 3 & 1 \end{bmatrix}$$

We compute all <u>upper-left</u> determinants:

$$D_1 = 2 \quad \oplus$$

$$D_2 = \begin{vmatrix} 2 & 0 \\ 0 & 8 \end{vmatrix} = 8$$

$$D_3 = \begin{vmatrix} 2 & 0 & 3 \\ 0 & 4 & 1 \\ 3 & 1 & 2 \end{vmatrix} = 2 \cdot \begin{vmatrix} 4 & 1 \\ 1 & 2 \end{vmatrix} + 3 \begin{vmatrix} 0 & 4 \\ 3 & 1 \end{vmatrix} = 2(7) + 3 \cdot (-12)$$

$$= -22 < 0 \quad \ominus$$

expand along the first row

<u>Conclusion:</u>

$D_3 < 0 \Rightarrow$ matrix A is not positive definite

e.g. Suppose, A is the **Hessian matrix** of a function $f(x,y)$ of two variables, <u>at a critical point</u> (x_0, y_0).

$$A = F(x_0, y_0) = \begin{bmatrix} f_{xx}(x_0,y_0) & f_{xy}(x_0,y_0) \\ f_{xy}(x_0,y_0) & f_{yy}(x_0,y_0) \end{bmatrix}$$

The upper left sub-determinants of this matrix are:

$$D_1 = f_{xx}(x_0, y_0)$$

$$D_2 = \begin{vmatrix} f_{xx}(x_0,y_0) & f_{xy}(x_0,y_0) \\ f_{xy}(x_0,y_0) & f_{yy}(x_0,y_0) \end{vmatrix} = f_{xx}(x_0,y_0) f_{yy}(x_0,y_0) - f_{xy}(x_0,y_0)^2$$

By Sylvester's criterion:

--- A is <u>positive definite</u> precisely when $D_1 > 0$ and $D_2 > 0$

--- A is <u>negative definite</u> precisely when $D_1 < 0$ and $D_2 > 0$

These are exactly the sufficient conditions for local min/max at the critical point (x_0, y_0) *!*

How to find the local max/min values of a function of several variables, $f(\mathbf{x}) = f(x_1, x_2, \ldots, x_n)$

(where all second order partial derivatives are continuous).

Theorem:

1. *(Necessary condition)* Find all the critical points $\mathbf{x}_0$ of f. These are points where $\nabla f(\mathbf{x}_0) = 0$.

(i.e. find all the points where **all** partial derivatives are zero)

A local extremum can only occur at a critical point.

2. *(Sufficient condition, "Second derivative test")*

Check the Hessian matrix at the critical points $\mathbf{x}_0$:

- If $F(\mathbf{x}_0) = \nabla^2 f(\mathbf{x}_0)$ is positive definite, then f has a local min at $\mathbf{x}_0$.
- If $F(\mathbf{x}_0) = \nabla^2 f(\mathbf{x}_0)$ is negative definite, then f has a local max at $\mathbf{x}_0$.

Example 7: Find the local max/min values of the function

Polynomial of degree 2 in variables x, y, z

$$f(\mathbf{x}) = f(x, y, z) = x^2 + 2x + 2y^2 - 8y + z^2 - 6z$$

Sol: <u>Step 1:</u> *(Necessary condition):*

$$\nabla f(\mathbf{x}) = 0 \qquad \text{means} \qquad f_x = 0, \quad f_y = 0, \quad f_z = 0.$$

(1) $f_x = 2x + 2 = 0 \quad \Rightarrow \quad 2x = -2 \Rightarrow \boxed{x = -1}$

(2) $f_y = 4y - 8 = 0 \quad \Rightarrow \quad 4y = 8 \Rightarrow \boxed{y = 2}$

(3) $f_z = 2z - 6 = 0 \quad \Rightarrow \quad 2z = 6 \Rightarrow \boxed{z = 3}$

There is <u>one</u> critical point: $\boxed{(x, y, z) = (-1, 2, 3).}$

<u>Step 2:</u> *(Sufficient condition)*: Check the Hessian matrix. By (1),(2),(3)

$f_{xx} = 2 \qquad f_{yy} = 4 \qquad f_{zz} = 2$

$f_{xy} = f_{yx} = 0 \qquad f_{xz} = f_{zx} = 0 \qquad f_{yz} = f_{zy} = 0$

$$F(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial x \partial z} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} & \frac{\partial^2 f}{\partial y \partial z} \\ \frac{\partial^2 f}{\partial z \partial x} & \frac{\partial^2 f}{\partial z \partial y} & \frac{\partial^2 f}{\partial z^2} \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$F(\vec{x})$ is constant, because the function $f(\vec{x})$ is a polynomial of degree 2.

At the critical point $(x, y, z) = (-1, 2, 3)$

$$F(-1,2,3) = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

This is a diagonal matrix. Because all diagonal entries are positive, then this matrix is <u>positive definite:</u> The function $f(\mathbf{x})$ has a local minimum at $\mathbf{x}^\star = (x, y, z) = (-1, 2, 3)$.

The local max value is:

$$f(-1, 2, 3) = (-1)^2 + 2(-1) + 2(2^2) - 8(2) + 3^2 - 6(3)$$

$$= 1 - 2 + 8 - 16 + 9 - 18 = -18$$

this local min is also an absolute min, See Example 9 below

IV.6.2 Newton's Method for finding critical points

It is often difficult to solve $\nabla f(\mathbf{x}) = 0$ analytically. In this case, one can use the generalization of Newton's method to several variables:

numerically

Recall Newton's method for finding critical points in one variable, i.e. solving $f'(x) = 0$, for $f : \mathbb{R} \to \mathbb{R}$. The iterations are

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$

$$x_{k+1} = x_k - f'(x_k)\left[f''(x_k)\right]^{-1}$$

$$x_{k+1} = x_k - \left[f''(x_k)\right]^{-1} f'(x_k)$$

To solve $\nabla f(\mathbf{x}) = 0$ for a function of several variables, $f : \mathbb{R}^n \to \mathbb{R}$, Newton's method becomes (depending on whether we write vectors in row form or in column form):

-- If we write vectors **in row form**, then

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \underbrace{\nabla f(\mathbf{x}_k)}_{\text{gradient}} \underbrace{[F(\mathbf{x}_k)]^{-1}}_{\text{inverse of Hessian matrix}} \qquad \text{(A)}$$

where

$F(\mathbf{x})$ is the Hessian matrix $\nabla^2 f(\mathbf{x})$

$[F(\mathbf{x})]^{-1}$ is the inverse of the Hessian matrix

-- If we write vectors **in column form**, then

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \underbrace{[F(\mathbf{x}_k)]^{-1}}_{\text{inverse of Hessian matrix}} \underbrace{\nabla f(\mathbf{x}_k)}_{\text{gradient}} \qquad \text{(B)}$$

In the following, we will use formula (B).

Example 8: Use Newton's method to find a critical point of

$$f(\mathbf{x}) = f(x,y) = 2x^2 - 2xy + y^2 + 2x - 2y$$

starting with $\mathbf{x}_0 = (0,0)$.

Sol: We agree to write all vectors in **column form:**

gradient as column vector

The gradient is

$$\nabla f(x,y) = \begin{bmatrix} f_x(x,y) \\ f_y(x,y) \end{bmatrix} = \begin{bmatrix} 4x - 2y + 2 \\ -2x + 2y - 2 \end{bmatrix}$$

The Hessian matrix is

$$\nabla^2 f(x,y) = F(x,y) = \begin{bmatrix} f_{xx}(x,y) & f_{yx}(x,y) \\ f_{xy}(x,y) & f_{yy}(x,y) \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ -2 & 2 \end{bmatrix}$$

the Hessian matrix is constant because f is a polynomial of degree 2 in x and y

<u>Recall:</u> If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $A^{-1} = \dfrac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Therefore,

$$\left[F(\vec{x})\right]^{-1} = \frac{1}{\underbrace{4\cdot 2-(-2)\cdot(-2)}_{=4}}\begin{bmatrix} 2 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix}$$

Start with $\vec{x_0} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ in formula (B):

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix}\begin{bmatrix} 4x_k - 2y_k - 2 \\ -2x_k + 2y_k + 2 \end{bmatrix}$$

$\underline{k=0:}$

$$\vec{x_1} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 4x_0 - 2y_0 - 2 \\ -2x_0 + 2y_0 + 2 \end{bmatrix}$$

$(x_0 = 0,\ y_0 = 0)$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{matrix} \leftarrow x_1 \\ \leftarrow y_1 \end{matrix}$$

$\underline{k=1:}$

$$\vec{x_2} = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 4x_1 - 2y_1 - 2 \\ -2x_1 + 2y_1 + 2 \end{bmatrix}$$

$x_1 = 0$
$y_1 = 1$

$\nabla f(\vec{x_1}) = 0$! This shows that $\vec{x_1}$ is a critical point

$$= \begin{bmatrix} 0 \\ 1 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \vec{x_1}$$

We see that $\vec{x_2} = \vec{x_1}$, so we stop.

[in general, we stop when $\vec{x}_{k+1}$ and $\vec{x}_k$ are very close to another]

The critical point is: $\vec{x}^* = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ (in vector form)

or $x^* = (0, 1)$ (as a point)

Next use the "second derivative test":

The Hessian matrix at x^* is

$$\nabla^2 f(\vec{x}^*) = F(\vec{x}^*) = \begin{bmatrix} 4 & -2 \\ -2 & 2 \end{bmatrix}$$ pos/neg. definite?

$D_1 = 4 > 0$, $\oplus$ $\qquad D_2 = \begin{vmatrix} 4 & -2 \\ -2 & 2 \end{vmatrix} = 8 - 4 = 4 > 0$ $\oplus$

D_1 and D_2 are both $\oplus$ $\Rightarrow$ $F(\vec{x}^*)$ is positive definite

$\Rightarrow$ f has a local <u>min</u> at $x^* = (0,1)$.

<u>Remark:</u> We could find x^* after only <u>one</u> iteration. This happens when f is a polynomial of degree ≤ 2.

Example 9: Redo Example 7,

$$f(\mathbf{x}) = f(x,y,z) \;=\; x^2 + 2x + 2y^2 - 8y + z^2 - 6z$$

but find the critical point by Newton's method, starting with $\mathbf{x}_0 = (0,1,1)$.

Sol: We have seen:

$$f_x(x,y,z) = 2x + 2, \qquad f_y(x,y,z) = 4y - 8, \qquad f_z(x,y,z) = 2z - 6\ ,$$

so the <u>gradient</u> (as column vector) is

$$\nabla f(x,y,z) \;=\; \begin{bmatrix} f_x \\ f_y \\ f_z \end{bmatrix} \;=\; \begin{bmatrix} 2x+2 \\ 4y-8 \\ 2z-6 \end{bmatrix}$$

We have seen: the Hessian matrix is

$$\nabla^2 f(\mathbf{x}) = F(\mathbf{x}) \;=\; \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Then the inverse of the Hessian matrix is $\quad [F(\mathbf{x})]^{-1} = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$

<u>Step 1:</u> Find the critical numbers: The Newton iterations are by (B),

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \left[F(\mathbf{x}_k)\right]^{-1} \nabla f(\mathbf{x}_k)$$

That is,

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \\ z_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \\ z_k \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2x_k + 2 \\ 4y_k - 8 \\ 2z_k - 6 \end{bmatrix}$$

Starting with $\mathbf{x}_0 = (0,1,1)$ in column form,

$$\mathbf{x}_0 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

k=0:

$$\mathbf{x}_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2x_0 + 2 \\ 4y_0 - 8 \\ 2z_0 - 6 \end{bmatrix} \begin{matrix} x_0 = 0 \\ x_0 = 1 \\ z_0 = 1 \end{matrix}$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 \\ -4 \\ -4 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} \begin{matrix} \leftarrow x_1 \\ \leftarrow y_1 \\ \leftarrow z_1 \end{matrix}$$

k=1: $$\mathbf{x}_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2x_1 + 2 \\ 4y_1 - 8 \\ 2z_1 - 6 \end{bmatrix}$$

$x_1 = -1$

$y_1 = 2$

$z_1 = 3$

$$= \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$$

We see that $\mathbf{x}_2 = \mathbf{x}_1$ already, so we stop. The critical point is (in row form)

$$\mathbf{x}^\star = (-1, 2, 3)$$ (which is the same as in Example 7)

<u>Step 2:</u> Because at the <u>critical point:</u>

1) The Hessian matrix $F(\mathbf{x}^\star)$ is of <u>diagonal</u> form
2) All the entries of $F(\mathbf{x}^\star)$ are <u>positive</u>

then $F(\mathbf{x}^\star)$ is positive definite.

$$\Rightarrow \quad \boxed{f \text{ has a local minimum at } \mathbf{x}^\star = (-1, 2, 3).}$$

IV.6.3 Examples of finding local min/max by Newton's Method

Example 12: Use Newton's method to find a critical point $\mathbf{x}^\star$ of

$$f(\mathbf{x}) = f(x,y,z,w) = (x+10y)^2 + 5(z-w)^2 + (y-2z)^4 + 10(x-w)^4$$

starting with $\mathbf{x}_0 = (3,-1,0,1)$. Then determine whether f has a local max. or min there.

Sol: We agree to write vectors in column form. The gradient and initial vetors are is

$$\nabla f(\mathbf{x}) = \begin{bmatrix} f_x \\ f_y \\ f_z \\ f_w \end{bmatrix} = \begin{bmatrix} 2(x+10y) + 40(x-w)^3 \\ 2(x+10y)\cdot 10 + 4(y-2z)^3 \\ 10(z-w) + 4(y-2z)^3(-2) \\ 10(z-w)(-1) + 40(x-w)^3(-1) \end{bmatrix} \qquad \mathbf{x}_0 = \begin{array}{c} \begin{bmatrix} 3 \\ -1 \\ 0 \\ 1 \end{bmatrix} \end{array} \begin{array}{l} -x_0 \\ -y_0 \\ -z_0 \\ -w_0 \end{array}$$

The Hessian matrix is

$$\nabla^2 f(\mathbf{x}) = F(\mathbf{x}) \;=\; \begin{bmatrix} f_{xx} & f_{yx} & f_{zx} & f_{wx} \\ f_{xy} & f_{yy} & f_{zy} & f_{wy} \\ f_{xz} & f_{yz} & f_{zz} & f_{wz} \\ f_{xw} & f_{yw} & f_{zw} & f_{ww} \end{bmatrix}$$

$$= \begin{bmatrix} 2+120(x-w)^2 & 20 & 0 & -120(x-w)^2 \\ 20 & 200+12(y-2z)^2 & -24(y-2z)^2 & 0 \\ 0 & -24(y-2z)^2 & 10+48(y-2z)^2 & -10 \\ -120(x-w)^2 & 0 & -10 & 10+120(x-w)^2 \end{bmatrix}$$

As usual, the Newton iterations are

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \left[F(\mathbf{x}_k)\right]^{-1} \nabla f\left(\mathbf{x}_k\right)$$

That is, (note the inverse sign)

$$\mathbf{x}_{k+1} = \begin{bmatrix} x_{k+1} \\ y_{k+1} \\ z_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \\ z_k \\ w_k \end{bmatrix} - \begin{bmatrix} 2+120(x_k-w_k)^2 & 20 & 0 & -120(x_k-w_k)^2 \\ 20 & 200+12(y_k-2z_k)^2 & -24(y_k-2z_k)^2 & 0 \\ 0 & -24(y_k-2z_k)^2 & 10+48(y_k-2z_k)^2 & -10 \\ -120(x_k-w_k)^2 & 0 & -10 & 10+120(x_k-w_k)^2 \end{bmatrix}^{-1} \begin{bmatrix} 2(x_k+10y_k)+40(x_k-w_k)^3 \\ 20(x_k+10y_k)+4(y_k-2z_k)^3 \\ 10(z_k-w_k)-8(y_k-2z_k)^3 \\ -10(z_k-w_k)-40(x_k-w_k)^3 \end{bmatrix}$$

By computer code, we obtain

```
Starting Point :   [ 3    -1    0    1 ]
 Iteration  0 : [ 1.587 -0.159 0.254 0.254 ]
 Iteration  1 : [ 1.058 -0.106 0.169 0.169 ]
 Iteration  2 : [ 0.705 -0.071 0.113 0.113 ]
 Iteration  3 : [ 0.470 -0.047 0.075 0.075 ]
 Iteration  4 : [ 0.314 -0.031 0.050 0.050 ]
 Iteration  5 : [ 0.209 -0.021 0.033 0.033 ]
 Iteration  6 : [ 0.139 -0.014 0.022 0.022 ]
 Iteration  7 : [ 0.093 -0.009 0.015 0.015 ]
 Iteration  8 : [ 0.062 -0.006 0.010 0.010 ]
 Iteration  9 : [ 0.041 -0.004 0.007 0.007 ]
Iteration  10 : [ 0.028 -0.003 0.004 0.004 ]
Iteration  11 : [ 0.018 -0.002 0.003 0.003 ]
Iteration  12 : [ 0.012 -0.001 0.002 0.002 ]
Iteration  13 : [ 0.008 -0.001 0.001 0.001 ]
Iteration  14 : [ 0.005 -0.001 0.001 0.001 ]
Iteration  15 : [ 0.004 -0.000 0.001 0.001 ]
Iteration  16 : [ 0.002 -0.000 0.000 0.000 ]
Iteration  17 : [ 0.002 -0.000 0.000 0.000 ]
Iteration  18 : [ 0.001 -0.000 0.000 0.000 ]
Iteration  19 : [ 0.001 -0.000 0.000 0.000 ]
Iteration  20 : [ 0.000 -0.000 0.000 0.000 ]
```

So $x^\star = (0,0,0,0)$. [as a point]

Next we check whether the Hessian matrix is positive definite at $x^\star$.

$$F(x^\star) = F(0,0,0,0) = \begin{bmatrix} 2 & 20 & 0 & 0 \\ 20 & 200 & 0 & 0 \\ 0 & 0 & 10 & -10 \\ 0 & 0 & -10 & 10 \end{bmatrix}$$

Compute the sub-determinants:

$D_1 = 2 > 0$ $\quad D_2 = \begin{vmatrix} 2 & 20 \\ 20 & 200 \end{vmatrix} = 0$!! $\Rightarrow$ $F(x^*)$ is not pos. definite !!

The second derivative test is inconclusive ! (When there are zeros on the diagonal of $F(x^\star)$, it may still be possible that max/min exist !)

Or when some sub-determinants are zero

Note:

$$D_3 = \begin{vmatrix} 2 & 20 & 0 \\ 2 & 200 & 0 \\ 0 & 0 & 10 \end{vmatrix} = 10 \cdot \begin{vmatrix} 2 & 20 \\ 20 & 200 \end{vmatrix}$$

$$= 10 \cdot 0 = 0 \; !!$$

$$D_4 = \left|\begin{array}{cc|cc} 2 & 20 & 0 & 0 \\ 20 & 200 & 0 & 0 \\ \hline 0 & 0 & 10 & -10 \\ 0 & 0 & -10 & 10 \end{array}\right| = \begin{vmatrix} 2 & 20 \\ 20 & 200 \end{vmatrix} \cdot \begin{vmatrix} 10 & -10 \\ -10 & 10 \end{vmatrix}$$

$$= 0 \cdot 0 = 0 \; !!$$

We can proceed by looking at formula of the function f.

$$f(x,y,z,w) = (x+10y)^2 + 5(z-w)^2 + (y-2z)^4 + 10(x-w)^4$$

We observe that $f(x,y,z,w)$

$(x,y,z,w) \geq 0$ for <u>all</u> x, y, z, w.

When $(x,y,z,w) = (0,0,0,0)$ then

$f(0,0,0,0) = 0$

$\Rightarrow$ f has its <u>global minimum</u> at the critical point $x^* = (0,0,0,0)$

The min. value is $f(0,0,0,0) = \underline{\underline{0}}$